the world, a number of islands including the **Galapagos island** and **Pacific island**. Darwin made elaborate observations on nature of the land, plants and animals of the regions he visited. He further worked for a period of 20 years to develop the theory of natural selection.

Darwin published his observations and conclusions under the name '**Origin of species**' in **1859**. The book of Darwin demonstrates the fact of evolution. It elaborates on the **theory of Natural selection** for evolutionary transformation,

### Principles of Darwinism

**i. Overproduction**

Living beings have the ability to reproduce more individuals and form their own progeny. They have the capacity to multiply in a geometrical manner. This will increase reproductive potential leading to overproduction.

**ii. Struggle for existence**

Due to over production, a geometric ratio of increase in population occurs. The space to live and food available for the organisms remain the same. This creates an intense competition among the organisms for food and space leading to struggle. The struggle for existence are of three types:

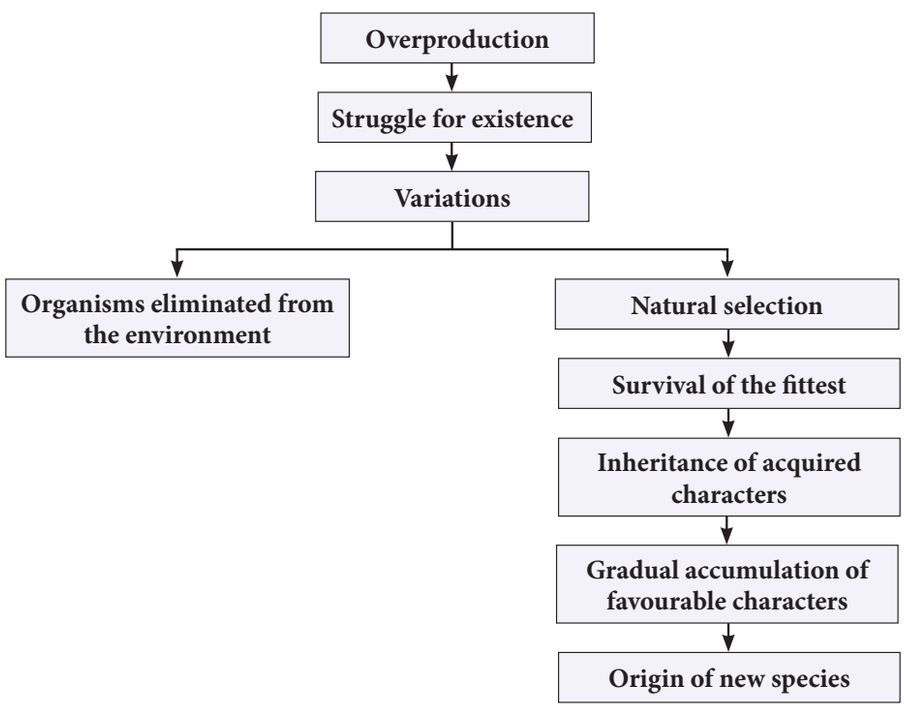
a. **Intraspecific struggle:** Competition among the individuals of same species.
b. **Interspecific struggle:** Competition between the organisms of different species living together.
c. **Environmental struggle:** Natural conditions like extreme heat or cold, drought and floods can affect the existence of organisms

**iii. Variations**

The occurrence of variation is a characteristic feature of all plants and animals. **Small variations** are important for evolution. According to Darwin **favourable variations** are useful to the organism and **unfavourable variations** are harmful or useless to the organism.

**iv. Survival of the fittest or Natural selection**

During the struggle for existence, the organisms which can overcome the challenging situation, **survive** and **adapt** to




**Flowchart showing the postulates of Darwinism**



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